

CSE 318 Assignment 4: Decision Tree Learning Analysis

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1 Introduction

This report presents the implementation and evaluation of a decision tree learning algorithm for the Iris and Adult datasets, as per the requirements of CSE 318 Assignment 4. The algorithm supports three attribute selection criteria: Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG). The program was tested with maximum tree depths from 0 to 6, using 20 random 80/20 train-test splits per configuration. This report includes graphs of average accuracy and node count versus maximum tree depth, and a comparison of when each criterion performs well or poorly for both datasets, reflecting the preprocessing applied to the Adult dataset.

2 Methodology

The decision tree algorithm was implemented in Java, accepting command-line parameters for the criterion (IG, IGR, NWIG) and maximum depth (maxDepth). The datasets are:

- **Iris:** 150 instances, 4 numerical attributes (discretized into quartiles), 3 classes (Iris-setosa, Iris-versicolor, Iris-virginica).
- **Adult:** 32,561 instances, 13 attributes (5 numerical discretized, 8 categorical, with fnlwgt removed), 2 classes ($\leq 50K$, $> 50K$).

Preprocessing in `DataHandler.java` includes:

- Discretizing numerical attributes (age, education-num, capital-gain, capital-loss, hours-per-week for Adult; all 4 for Iris) into quartiles (Q1, Q2, Q3, Q4).
- Mapping missing values (?) in Adult to Unknown.
- Grouping high-cardinality attributes in Adult:
 - native-country: Grouped into North-America, Asia, Europe, South-America, Other.
 - occupation: Grouped into Professional, Blue-collar, Service, Sales, Management, Administrative, Military, Other.
 - workclass: Grouped into Private, Self-employed, Government, Other.

The criteria are defined as:

- **IG:** $IG(S,A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \cdot Entropy(S_v)$
- **IGR:** $GainRatio(S,A) = \frac{IG(S,A)}{IV(A)}$, where $IV(A) = -\sum_{v \in Values(A)} \frac{|S_v|}{|S|} \cdot \log_2 \left(\frac{|S_v|}{|S|} \right)$
- **NWIG:** $NWIG(S,A) = \left(\frac{IG(S,A)}{\log_2(k+1)} \right) \cdot \left(1 - \frac{k-1}{|S|} \right)$

For each dataset, criterion, and depth, 20 experiments were conducted, and average accuracy, node count, and tree depth were recorded.

3 Results

Tables 1 and 2 summarize the results for the Iris and Adult datasets, respectively. Figures 1, 2, 3, and 4 plot average accuracy and node count versus maximum tree depth.

Table 1: Results for Iris Dataset

Criterion	MaxDepth	Avg Accuracy (%)	Avg Node Count	Avg Tree Depth
IG	0	92.9032	23.00	3.80
IG	1	84.5161	5.00	1.00
IG	2	95.8065	13.00	2.00
IG	3	93.3871	19.20	3.00
IG	4	94.8387	24.80	3.85
IG	5	92.5806	22.80	3.75
IG	6	94.8387	24.60	3.95
IGR	0	94.8387	24.60	3.85
IGR	1	83.7097	5.00	1.00
IGR	2	95.1613	13.00	2.00
IGR	3	94.1935	19.20	3.00
IGR	4	93.3871	24.00	3.95
IGR	5	93.0645	23.00	3.85
IGR	6	93.0645	24.00	3.95
NWIG	0	93.0645	23.80	3.90
NWIG	1	83.3871	5.00	1.00
NWIG	2	95.9677	13.00	2.00
NWIG	3	94.6774	19.20	3.00
NWIG	4	94.5161	24.60	3.85
NWIG	5	95.1613	25.00	3.85
NWIG	6	94.1935	24.80	3.95

Figure 1: Average Accuracy vs. Max Tree Depth for Iris Dataset

Table 2: Results for Adult Dataset

Criterion	MaxDepth	Avg Accuracy (%)	Avg Node Count	Avg Tree Depth
IG	0	81.7764	37366.40	13.00
IG	1	75.7424	8.00	1.00
IG	2	81.8970	89.45	2.00
IG	3	82.2877	528.50	3.00
IG	4	82.6524	1602.45	4.00
IG	5	82.7315	3503.35	5.00
IG	6	82.5902	6163.90	6.00
IGR	0	81.6613	29040.65	13.00
IGR	1	77.7476	5.00	1.00
IGR	2	78.4823	20.25	2.00
IGR	3	80.8038	92.75	3.00
IGR	4	83.1199	295.00	4.00
IGR	5	83.1898	843.25	5.00
IGR	6	83.2635	1944.30	6.00
NWIG	0	81.9807	51779.25	13.00
NWIG	1	76.0088	8.00	1.00
NWIG	2	82.0643	36.00	2.00
NWIG	3	82.3376	158.85	3.00
NWIG	4	82.6800	582.90	4.00
NWIG	5	83.1905	1683.40	5.00
NWIG	6	83.2497	3950.25	6.00

Figure 2: Average Node Count vs. Max Tree Depth for Iris Dataset

4 Observations and Analysis

The results highlight when the decision tree algorithm performs well or poorly for the Iris and Adult datasets across the criteria (IG, IGR, NWIG) and maximum tree depths (0 to 6). The Adult dataset benefits from grouping high-cardinality attributes (native-country, occupation, workclass), reducing tree complexity while maintaining performance.

4.1 Iris Dataset

- **When It Works Well:**

- NWIG at maxDepth = 2 achieves the highest accuracy (95.97%), followed closely by IG (95.81%) and IGR (95.16%) at the same depth. This depth balances tree complexity (13.00 nodes) with high accuracy.
- maxDepth = 4–6 yields consistently high accuracies (93.06%–94.84% for IG and IGR; 94.19%–95.16% for NWIG), indicating robust performance due to Iris’s small size (150 instances) and clear class separation (3 classes).
- IGR at maxDepth = 0 (94.84%, 24.60 nodes) performs comparably to higher depths, suggesting the datasets simplicity allows effective trees without depth constraints.

Figure 3: Average Accuracy vs. Max Tree Depth for Adult Dataset

Figure 4: Average Node Count vs. Max Tree Depth for Adult Dataset

- **When It Works Poorly:**

- All criteria perform worst at `maxDepth = 1` (IG: 84.52%, IGR: 83.71%, NWIG: 83.39%), with minimal node counts (5.00), as the shallow depth limits the trees ability to capture class distinctions.
- IG at `maxDepth = 5` (92.58%) and IGR at `maxDepth = 5-6` (93.06%) show slightly lower accuracies compared to `maxDepth = 2`, indicating deeper trees add unnecessary complexity for Iris.

4.2 Adult Dataset

- **When It Works Well:**

- IGR and NWIG perform best at `maxDepth = 4-6`, with IGR achieving 83.26% at `maxDepth = 6` (1944.30 nodes) and NWIG at 83.25% (3950.25 nodes). These depths allow the tree to leverage the 13 attributes effectively, enhanced by grouping high-cardinality attributes.
- NWIG at `maxDepth = 0` (81.98%, 51779.25 nodes) outperforms IG (81.78%) and IGR (81.66%), benefiting from its normalization that mitigates bias toward high-cardinality attributes, even with grouped categories.
- `maxDepth = 3` yields good performance (IG: 82.29%, NWIG: 82.34%, IGR: 80.80%) with moderate node counts (92.75–528.50), showing the grouped attributes reduce tree size while maintaining accuracy.

- **When It Works Poorly:**

- All criteria perform worst at `maxDepth = 1` (IG: 75.74%, NWIG: 76.01%, IGR: 77.75%) with very few nodes (5.00–8.00), as the shallow depth cannot capture the complexity of 13 attributes and 32,561 instances.
- IGR at `maxDepth = 2` (78.48%, 20.25 nodes) is notably worse than IG (81.90%) and NWIG (82.06%) at the same depth, suggesting its penalty for attribute splits is too restrictive for shallow trees.
- IG at `maxDepth = 0` (81.78%, 37366.40 nodes) is slightly worse than NWIG, as it may select high-cardinality attributes (even after grouping) that lead to larger, less effective trees.

4.3 Impact of Preprocessing on Adult

The grouping of high-cardinality attributes (`native-country` into 5 regions, `occupation` into 8 categories, `workclass` into 4 types) significantly reduces node counts compared to ungrouped data. For example, at `maxDepth = 3`, IG yields 528.50 nodes, much lower than

expected without grouping (e.g., native-country with 40+ values). This preprocessing improves computational efficiency and maintains high accuracy, particularly for IGR and NWIG at `maxDepth = 4-6`.

5 Conclusion

The decision tree algorithm performs robustly on the Iris dataset, with NWIG at `maxDepth = 2` achieving the highest accuracy (95.97%) and all criteria performing well at `maxDepth  $\geq$  2`. For the Adult dataset, IGR and NWIG excel at `maxDepth = 4-6` (83.19%–83.26%), benefiting from grouped attributes that reduce tree complexity. The algorithm struggles at `maxDepth = 1` for both datasets and for IGR at `maxDepth = 2` on Adult. The preprocessing of high-cardinality attributes in Adult enhances efficiency and performance, making IGR and NWIG preferable for complex datasets.